



feature

JAXA launches Ibuki

Launched on 23 Jan 2009, Ibuki is the first satellite for collecting data on greenhouse gas emissions

mission to the ISS scheduled for 13 March 2009. By the time this issue rolls out, you should have seen plenty of coverage of the Discovery mission delivering components to the ISS. The primary purpose of the mission is to deliver solar panels to the space station. A number of experiments will also be carried out. The shuttle will carry a crew of nine. Later in the same month, on 26 March, RKA's Soyuz TMA-14 mission will take two cosmonauts and a space tourist to the ISS.

The space tourist in question is a Hungarian named Charles Simonyi. He previously went into space in 2007, as a space tourist to the ISS. Interestingly, Simonyi worked on our

dearly beloved MS Office and has managed to bag the privilege of being the last space tourist to go aboard the ISS for some time, because the shuttles will be phased out in the near future. This means that only private crafts and Soyuz will service the ISS, leaving very little room for space tourists. Later, on 23 May 2009, the Atlantis Space Shuttle will carry a crew of seven for a final servicing mission to the dying Hubble Space telescope — STS-125. The purpose of the mission will be to install some guidance systems and gyroscopes on the telescope, which will let it continue working for a while. This will be an interesting mission for all of us, as an IMAX camera will be taken into space for the second time, to film a documentary on the Hubble telescope. The STS-400 is a Shuttle mission to watch out for. Due to the age of the space shuttles, surface damage to the shuttle poses a considerable danger. In case any problems develop with the Atlantis mission, Endeavor will launch close on its heels and rescue the astronauts: a first of its kind shuttle-to-shuttle transfer.

Sometime in the same month, the RSA will launch the Soyuz TMA-15, a mission to transport crew to the ISS. Aboard this Russian craft, will be one member from each of Belgium and Canada. Apart from the contingency mission of STS-400, Endeavor will be on schedule for a May launch to the ISS, the STS-127. The most important component being delivered to the space station will be the Japanese Kibo laboratory. This mission requires extensive spacewalks by the people on board, and is bound to produce some interesting images. From this mission onwards, there will always be a six member crew on the ISS.

The Soyuz TMA-16 launch will take place some time in September this year. The three man crew will consist of an American, a Russian and a Kazakh cosmonaut. The Soyuz TMA-17 mission will reach the ISS in November. The crew will comprise a Japanese, an American and a Russian. The Atlantis STS-129 mission is scheduled sometime in November this year. This mission will carry an assortment of parts for the ISS. A large portion of these parts will be spares, only to be used when needed. The final human spaceflight planned this year is in mid-December. The STS-130 will use the space shuttle Endeavor. This will be an important mission for the station, as the robotics control centre will be launched in the payload. This module will control the robotic arm of the ISS, along with other deployed robotics and robotic attachments to follow the next year.

2010

The first flight planned for 2010 will be the STS-131 mission, due in for February. This will be the final mission for the short-lived, but incredibly useful MPLM (see box). This mission will deliver



Launch vehicles around the world. From left to right, the Soyuz (RSA), the Saturn V (NASA), the GSLV (ISRO), the H2A204 (JAXA), the LM2E (CNSA), the Ariane 5 (ESA)

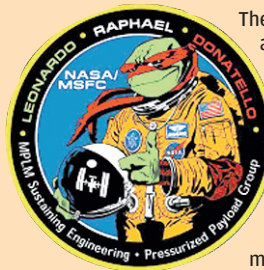
a spare ammonia tank to the ISS, along with a sleeping area to accommodate more crew. This shuttle mission will also attach a logistics module. The space shuttle Atlantis will be used for the final journey during this mission. The STS-132 is scheduled later in April. This will be the final flight of the Discovery shuttle. A mini-research Russian module will be sent to the ISS, along with Canadian and Japanese robotic arms. The Soyuz-TMA 18 launch will send three cosmonauts to the ISS in May. At the end of May, the Soyuz-TMA-01M launch will use the new class of Soyuz craft.

The STS-133 mission due in May 2010 will be historic for several reasons.

This shuttle mission will deliver components of the ISS, the EXPRESS Logistics Carrier. Although more components will be added later on, the ISS will be said to be completed on this mission. The space shuttle Endeavor will fly for the last time, after which the shuttles will retire, completing 33 years of service. Towards the end of September, the Soyuz-TMA-01M launch will use the new class of Soyuz craft. These spacecraft, known as the Soyuz TMA-T have a number of upgrades including a new parachuting system for re-entry, a glass cockpit, and upgraded electronics. There are some other changes that allow better docking with the ISS and better compatibility with the American counterparts.

However, the next Soyuz mission in November — the TMA-20, will use the old Soyuz spacecraft. NASA's Commercial Orbital Transportation Service (COTS) program will take over after the retirement of the shuttles. These are contracts given out to private organisations to supply and man the ISS. The Taurus II, a launch vehicle built by the Orbital Sciences Corporation will carry the unmanned Cygnus spacecraft into a test orbit, and possibly re-supply the ISS in the later part of 2010. Although

Multi Purpose Logistics Module (MPLM)



The Multi Purpose Logistics Module is a hybrid between a cargo bay and a spacecraft. There are three modules in existence, constructed by the Italian Space Agency. The modules are named Donatello, Raffaello and Leonardo, after the famous Italian talents. Raffaello and Leonardo are being used repeatedly on a number of Shuttle flights. The modules are mounted on the shuttles during transportation. Once at the ISS, the

modules can be detached from the shuttles and docked with the space station itself. At this juncture, the modules support humans because they are pressurised and have certain vital systems such as life support and fire alarms. The module normally stores additional equipment for the station, and returns back to Earth with the shuttles. When piggybacking on the shuttle, humans cannot enter or exit the module. The Donatello module is a more advanced version, and has not been used as yet, and is not likely to be used before the shuttles are retired.